V-01: Calcium-associated reporter schematic

How can the reporter principle be explained without claiming a molecular trajectory?

Materials and methods

Existing three-panel storyboard; Chen et al. (2013) primary GCaMP6 study.

Draw cpGFP, CaM and M13 as labelled conceptual domains. Show calcium-associated CaM-M13 interaction and emission under excitation.

Export original SVG/PNG artwork and a looping GIF using the storyboard's presentation durations.

Scope and working assumptions

The artwork is a conceptual GCaMP-family explanation based on the supplied storyboard and primary reference. Domain geometry, calcium-symbol count, brightness and timing are presentation choices. The visual illustrates calcium-associated reporting, rather than a molecular trajectory or a measured response of the tested clone.

Results

Created 3 panels and an animation lasting 12 presentation seconds per loop.

Output	Format	Scope	Input data	Finished without more wet-lab data?
Storyboard	PNG + SVG	Editable original schematic	Storyboard	Yes: conceptual
Animation	GIF	Three presentation states	Storyboard	Yes: conceptual

Interpretation

The completed visual presents calcium-associated CaM-M13 interaction and reporter emission in three panels. Editable SVG and PNG files accompany a GIF with a 12-second presentation loop.

References and reproducibility

Chen et al. (2013), Ultrasensitive fluorescent proteins for imaging neuronal activity. https://doi.org/10.1038/nature12354

Complete inline code: tasks/V-01/V-01.ipynb. Numerical data and source/software provenance: data/output/V-01/.

Figures

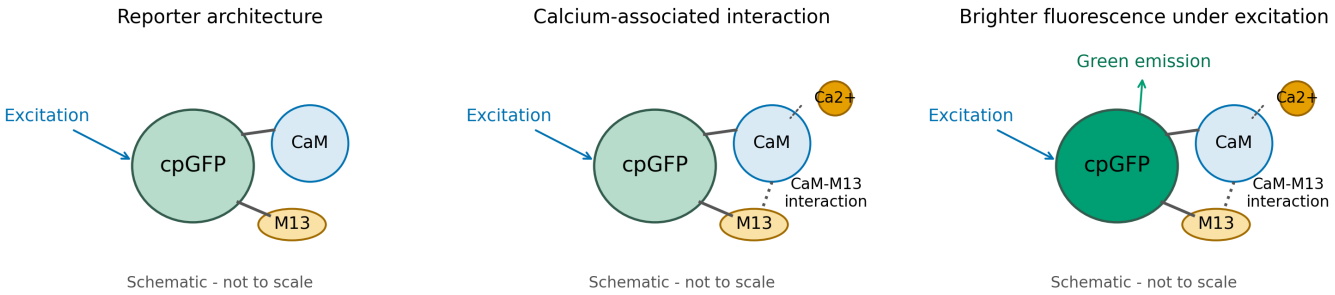


Figure 1. Original conceptual GCaMP-family storyboard. Domain geometry, calcium-symbol count, brightness and animation timing are illustrative, not structural or kinetic results.